Time and event ordering

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Sample chapter for the proposed course plan.

1 Learning outcome

Distinguish causal order from wall-clock timestamp order.

Prerequisites: Lectures 01 and 03.

2 Core idea

Different machines can report different wall-clock times for related events. Message exchange gives another ordering rule: a message must be sent before that same message is received. This helps reason about causality without assuming perfectly synchronized clocks.

3 Worked scenario

The registration service sends a confirmation event. A notification worker receives it and sends an email. The worker clock happens to be behind the service clock.

4 Discussion

Can the email have an earlier timestamp without having caused the registration?

5 Recap

Reason about causal dependencies before interpreting timestamps.